Quantum dynamics

Week

III

In the previous week, we reached the point where our optical problem became quantum and dynamical. Indeed, we state that Maxwell's equations tell us how an electromagnetic field propagates once the material polarization is known. To construct this polarization, the semiclassical description tells us that this polarization is obtained from a quantum expectation value of the electric dipole operator for identical dipoles of number density $\mathcal{N}$,

$$\boldsymbol{P}(\boldsymbol{r},t) = \mathcal{N}\,\langle\psi(t)|\hat{\boldsymbol{p}}(t)|\psi(t)\rangle = \mathcal{N}\,\langle\hat{\boldsymbol{p}}(t)\rangle. \tag{0.1}$$

The remaining question is to understand how the state of the material changes in time when an electromagnetic field is applied. A particularly useful way of thinking about the semiclassical problem is

$$\boxed{\boldsymbol{E}(t) \longrightarrow \hat{H}(t) \longrightarrow |\psi(t)\rangle \longrightarrow \langle\hat{\boldsymbol{p}}(t)\rangle \longrightarrow \boldsymbol{P}(t) \longrightarrow \boldsymbol{E}(t).} \tag{0.2}$$

which means that the electromagnetic field enters the quantum Hamiltonian as a time-dependent drive. The quantum state responds to that drive, producing a time-dependent dipole moment. The ensemble of dipoles gives the macroscopic polarization, which then enters Maxwell's equations again. In a fully self-consistent treatment, these two dynamical problems are coupled, which makes the full dynamics of this loop a mathematically challenging problem. Nevertheless, interesting insights are still possible if we tackle it in a perturbative manner, which will be our starting point this week. We will then solve the near-resonant two-level dynamics within the rotating-wave approximation, still without decay.

1 Brief historical perspective and state representations

Before discussing time dynamics of quantum states in detail, we should mention that the first formulations of quantum mechanics did not arrive as a single mathematical language. In the mid-1920s Heisenberg developed a formulation based directly on measurable transition quantities, which Born and Jordan recognized as a matrix algebra. Shortly afterwards, Schrödinger introduced wave mechanics, describing the state of a quantum system by a

wavefunction satisfying a differential equation. The two approaches looked very different: one was built from matrices and non-commuting quantities, the other from waves and differential operators. It was soon understood that they were not competing theories and actually different representations of the same quantum mechanics. Dirac's transformation theory made this viewpoint particularly transparent, associating the physical state to an abstract vector in Hilbert space and the wavefunction or column vector to a representation in a chosen basis. The same is true for observables, which may appear as differential operators in one representation and as matrices in another.

The key point is that the same freedom also appears in the description of time evolution. In Schrödinger's formulation the state carries the time dependence. In Heisenberg's formulation the state is fixed, and the operators evolve. A third formulation, now usually called the interaction picture[1], shares the time dependence between states and operators in a way that is especially useful when a known system is driven by an external perturbation. This freedom is one of the main practical tools of quantum optics to turn complicated mathematical problems into simpler ones, as we shall explore in practice in the next few weeks.

[1] *The interaction picture is also frequently called the Dirac picture.*

1.1 Vector and matrix representations

Before exploring in detail the topic of quantum dynamics of the wavefunction itself, we will discuss briefly how we can represent states and operators as vectors and matrices, which in the context of quantum optics is very useful to simplify calculations. To exemplify, we take the ordered basis of a two-level system $\{|1\rangle, |2\rangle\}$. The state may be represented by the column vector

$$|\psi\rangle = c_1|1\rangle + c_2|2\rangle \longleftrightarrow \begin{pmatrix} c_1 \\ c_2 \end{pmatrix}. \tag{1.1}$$

In this vector representation, the operator $\hat{O}$ becomes

$$\hat{O} \longleftrightarrow \begin{pmatrix} O_{11} & O_{12} \\ O_{21} & O_{22} \end{pmatrix}, \text{ where } O_{mn} = \langle m|\hat{O}|n\rangle. \tag{1.2}$$

Put in this way, the diagonal matrix elements describe the action of the operator within each state, whereas the off-diagonal elements describe coupling between different basis states. Specifically, for the Hamiltonian,

$$\hat{H} \longleftrightarrow \begin{pmatrix} H_{11} & H_{12} \\ H_{21} & H_{22} \end{pmatrix}, \tag{1.3}$$

the off-diagonal terms will therefore be the terms that are capable of transferring probability amplitude between the two states.

1.2 Hamiltonian of a two-level atom

Atoms and solids generally have many energy levels, and in principle the state may require a large number of amplitudes. But, in some conditions, we can isolate a much smaller subspace, which is what happens in near-resonant phenomena. Indeed, if the applied field is spectrally narrow and close to one allowed transition, levels that are far from resonance or weakly connected by the relevant dipole operator may contribute only weakly to the dynamics. This is the basis of the **two-level toy model**[2]. Retaining only two states, say

$$\hat{H}_0|1\rangle = E_1|1\rangle, \qquad \hat{H}_0|2\rangle = E_2|2\rangle, \tag{1.4}$$

with $E_2 > E_1$, the wavefunction is simply given by

$$|\psi(t)\rangle = c_1(t)|1\rangle + c_2(t)|2\rangle, \qquad |c_1|^2 + |c_2|^2 = 1. \tag{1.5}$$

[2] *The approximation does not mean that the material literally possesses only two states. It means that, for the process and timescale of interest, the relevant dynamics are well represented in this two-dimensional subspace.*

In the ordered basis $\{|1\rangle, |2\rangle\}$ the bare Hamiltonian is

$$\hat{H}_0 = E_1|1\rangle\langle 1| + E_2|2\rangle\langle 2| \longleftrightarrow \begin{pmatrix} E_1 & 0 \\ 0 & E_2 \end{pmatrix}. \tag{1.6}$$

Defining the transition frequency $\omega_0 = (E_2 - E_1)/\hbar > 0$ and introducing the Pauli matrices,

$$\hat{\sigma}_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \hat{\sigma}_y = \begin{pmatrix} 0 & i \\ -i & 0 \end{pmatrix}, \quad \hat{\sigma}_z = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}. \tag{1.7}$$

we have that

$$\hat{\sigma}_z = |2\rangle\langle 2| - |1\rangle\langle 1|, \qquad \hat{\sigma}_+ = |2\rangle\langle 1|, \quad \hat{\sigma}_- = |1\rangle\langle 2|. \tag{1.8}$$

defining $\hat{\sigma}_\pm = (\hat{\sigma}_x \pm i\hat{\sigma}_y)/2$ and noting that $[\hat{\sigma}_i, \hat{\sigma}_j] = 2i\epsilon_{ijk}\hat{\sigma}_k$. Direct substitution yields

$$\hat{H}_0 = \frac{E_1 + E_2}{2}\hat{\mathbb{I}} + \frac{\hbar\omega_0}{2}\hat{\sigma}_z. \tag{1.9}$$

or simplifying by removing the energy shift that produces only a global phase, gives

$$\boxed{\hat{H}_0' = \frac{\hbar\omega_0}{2}\hat{\sigma}_z = \begin{pmatrix} -\frac{\hbar\omega_0}{2} & 0 \\ 0 & \frac{\hbar\omega_0}{2} \end{pmatrix}.} \tag{1.10}$$

1.3 The atom in an electromagnetic field

The following derivation follows the sequence and argument of Rand's book[3]. We begin with the classical force and construct the Hamiltonian before quantizing it. For consistency of notation, we write the canonical momentum as $\boldsymbol{\pi}$, reserving $\boldsymbol{p}$ for the electric dipole moment. The nonrelativistic force on a charged particle of velocity $\boldsymbol{v}$ is the Lorentz force,

[3] *Rand, Stephen Colby. Lectures on light: nonlinear and quantum optics using the density matrix. Oxford University Press, 2016, Appendix C*

$$\boxed{\boldsymbol{F} = q\left(\boldsymbol{E} + \boldsymbol{v} \times \boldsymbol{B}\right).} \tag{1.11}$$

To attack this definition and connect with the Hamiltonian formalism, we need to connect with a potential picture for the electromagnetic field. On the basis of the vector identity

$$\boldsymbol{\nabla} \cdot (\boldsymbol{\nabla} \times \boldsymbol{A}) = 0, \tag{1.12}$$

we may express a magnetic field satisfying Maxwell's equation $\boldsymbol{\nabla} \cdot \boldsymbol{B} = 0$ locally in the form

$$\boldsymbol{B} = \boldsymbol{\nabla} \times \boldsymbol{A}, \tag{1.13}$$

where $\boldsymbol{A}$ is called **a vector potential**. Similarly, applying Eq. 1.13 in Maxwell's equation $\boldsymbol{\nabla} \times \boldsymbol{E} = -\partial \boldsymbol{B}/\partial t$, we find

$$\boldsymbol{\nabla} \times \boldsymbol{E} + \frac{\partial}{\partial t}(\boldsymbol{\nabla} \times \boldsymbol{A}) = 0. \tag{1.14}$$

Reversing the order of the spatial and temporal derivatives gives

$$\boldsymbol{\nabla} \times \left(\boldsymbol{E} + \frac{\partial \boldsymbol{A}}{\partial t}\right) = 0. \tag{1.15}$$

Making use of a second vector identity, $\boldsymbol{\nabla} \times \boldsymbol{\nabla}\phi = 0$, where ϕ is a **scalar potential**, we may write the electric field as

$$\boxed{\boldsymbol{E} = -\boldsymbol{\nabla}\phi - \frac{\partial \boldsymbol{A}}{\partial t}.} \tag{1.16}$$

The expression for the Lorentz force becomes

$$\boldsymbol{F} = q\left[-\boldsymbol{\nabla}\phi - \frac{\partial \boldsymbol{A}}{\partial t} + \boldsymbol{v} \times (\boldsymbol{\nabla} \times \boldsymbol{A})\right]. \tag{1.17}$$

Now we shall work a bit with vector calculus to simplify this expression. First, we note that along the particle trajectory, the total derivative of the vector potential is

$$\frac{d\boldsymbol{A}}{dt} = \frac{\partial \boldsymbol{A}}{\partial t} + (\boldsymbol{v} \cdot \boldsymbol{\nabla})\boldsymbol{A}. \tag{1.18}$$

Also, taking spatial derivatives at fixed $\boldsymbol{v}$,

$$\boldsymbol{v} \times (\boldsymbol{\nabla} \times \boldsymbol{A}) = \boldsymbol{\nabla}(\boldsymbol{v} \cdot \boldsymbol{A}) - (\boldsymbol{v} \cdot \boldsymbol{\nabla})\boldsymbol{A}. \tag{1.19}$$

Using these replacements, we find, step by step,

$$\boldsymbol{F} = q\left[-\boldsymbol{\nabla}\phi + \boldsymbol{\nabla}(\boldsymbol{v} \cdot \boldsymbol{A}) - \frac{\partial \boldsymbol{A}}{\partial t} - (\boldsymbol{v} \cdot \boldsymbol{\nabla})\boldsymbol{A}\right] \tag{1.20}$$

We may now introduce the generalized, velocity-dependent potential

$$U(\boldsymbol{r}, \boldsymbol{v}, t) \equiv q\left(\phi - \boldsymbol{A} \cdot \boldsymbol{v}\right), \tag{1.21}$$

noting that the ith component of the force can be written as

$$\boxed{F_i = -\frac{\partial U}{\partial x_i} + \frac{d}{dt}\frac{\partial U}{\partial v_i}, \qquad i = x, y, z.} \tag{1.22}$$

Indeed, $\partial U/\partial v_i = -qA_i$, so the second term is $-q\, dA_i/dt$, precisely as in Eq. 1.20. The charge is already included in U. This generalized potential can then be used in the Lagrangian of nonrelativistic mechanics:

$$\boxed{L = T - U = \frac{1}{2}mv^2 - q\phi + q\boldsymbol{A} \cdot \boldsymbol{v}.} \tag{1.23}$$

[4] and in the construction of the Hamiltonian by a Legendre transform,

$$\begin{aligned} H &= \sum_i v_i \frac{\partial L}{\partial v_i} - L = \boldsymbol{v} \cdot (m\boldsymbol{v} + q\boldsymbol{A}) - \left(\frac{1}{2}mv^2 - q\phi + q\boldsymbol{A} \cdot \boldsymbol{v}\right) \\ &= \frac{1}{2}mv^2 + q\phi. \end{aligned} \tag{1.25}$$

However, for a correct first quantization procedure, the independent variables of a Hamiltonian are position and *canonical momentum*, not position and velocity. Indeed, in this case, the canonical momentum is

$$\pi_i = \frac{\partial L}{\partial v_i} = mv_i + qA_i \quad \Rightarrow \quad \boldsymbol{v} = \frac{\boldsymbol{\pi} - q\boldsymbol{A}}{m}. \tag{1.26}$$

Expressing Eq. 1.25 in these variables yields

$$\boxed{H = \frac{1}{2m}(\boldsymbol{\pi} - q\boldsymbol{A})^2 + q\phi.} \tag{1.27}$$

which corresponds to the **minimal-coupling Hamiltonian**. For a bound atomic particle, we also add the field-free binding potential energy $U_{\text{bind}}(\boldsymbol{r})$;

[4] *following the Euler–Lagrange equation*

$$\frac{d}{dt}\frac{\partial L}{\partial v_i} - \frac{\partial L}{\partial x_i} = 0 \tag{1.24}$$

it is easy to show that the Lorentz force is recovered.

the potentials ϕ and $\boldsymbol{A}$ below describe the applied field.

Having the Hamiltonian, is now time do go back and convert to a form expressed in terms of the applied fields $\boldsymbol{E}$ and $\boldsymbol{B}$, separating their multipole contributions. For this, we choose the origin at the atomic centre and a multipolar gauge in which the potentials have the expansions

$$\phi(\boldsymbol{r},t) = -\boldsymbol{r}\cdot\boldsymbol{E}(\boldsymbol{0},t) - \frac{1}{2}\sum_{i,j} r_i r_j\, \partial_j E_i(\boldsymbol{0},t) + \cdots , \tag{1.28}$$

$$\boldsymbol{A}(\boldsymbol{r},t) = \frac{1}{2}\boldsymbol{B}(\boldsymbol{0},t)\times\boldsymbol{r} + \frac{1}{3}\left[(\boldsymbol{r}\cdot\boldsymbol{\nabla})\boldsymbol{B}(\boldsymbol{0},t)\right]\times\boldsymbol{r} + \cdots . \tag{1.29}$$

These expansions constitute a specific choice of gauge, consistent with the Taylor expansions of the fields themselves,

$$\boldsymbol{E}(\boldsymbol{r},t) = \boldsymbol{E}(\boldsymbol{0},t) + (\boldsymbol{r}\cdot\boldsymbol{\nabla})\boldsymbol{E}(\boldsymbol{0},t) + \cdots , \tag{1.30}$$

$$\boldsymbol{B}(\boldsymbol{r},t) = \boldsymbol{B}(\boldsymbol{0},t) + (\boldsymbol{r}\cdot\boldsymbol{\nabla})\boldsymbol{B}(\boldsymbol{0},t) + \cdots . \tag{1.31}$$

with field derivatives evaluated at the origin. If the field varies little across an atom of size a, so that $ka \ll 1$ (i.e. the wavelength of the field is much larger than atom size - **dipolar approximation**), we keep only the leading terms that give

$$\phi \simeq -\boldsymbol{r}\cdot\boldsymbol{E}(\boldsymbol{0},t), \qquad \boldsymbol{A} \simeq \frac{1}{2}\boldsymbol{B}(\boldsymbol{0},t)\times\boldsymbol{r}. \tag{1.32}$$

We can now make the substitution into the Hamiltonian explicitly. First expand the classical square (introducing the U_{bind} potential that warrants stability of the atom),

$$H = \underbrace{\frac{\pi^2}{2m} + U_{\text{bind}}}_{H_0} + q\phi - \frac{q}{m}\boldsymbol{\pi}\cdot\boldsymbol{A} + \frac{q^2}{2m}A^2. \tag{1.33}$$

The electric term becomes

$$q\phi \simeq -q\boldsymbol{r}\cdot\boldsymbol{E} = -\boldsymbol{\mu}^{(e)}\cdot\boldsymbol{E}, \qquad \boldsymbol{\mu}^{(e)} = q\boldsymbol{r}. \tag{1.34}$$

For the term linear in $\boldsymbol{A}$, the scalar triple-product identity gives

$$\begin{aligned} -\frac{q}{m}\boldsymbol{\pi}\cdot\boldsymbol{A} &\simeq -\frac{q}{2m}\boldsymbol{\pi}\cdot(\boldsymbol{B}\times\boldsymbol{r}) \\ &= -\frac{q}{2m}(\boldsymbol{r}\times\boldsymbol{\pi})\cdot\boldsymbol{B} = -\boldsymbol{\mu}^{(m)}\cdot\boldsymbol{B}, \end{aligned} \tag{1.35}$$

where $\boldsymbol{\mu}^{(m)} = q(\boldsymbol{r}\times\boldsymbol{\pi})/(2m)$ is the orbital magnetic moment. To the retained order, it is also $\frac{1}{2}q\boldsymbol{r}\times\boldsymbol{v}$. The A^2 term is quadratic in the magnetic field

and gives a diamagnetic contribution, and it is safe to neglect it and the higher multipoles, retaining only the leading electric and magnetic dipole interactions. Thus the classical *interaction* Hamiltonian is

$$\boxed{V = -\boldsymbol{\mu}^{(e)} \cdot \boldsymbol{E} - \boldsymbol{\mu}^{(m)} \cdot \boldsymbol{B}, \qquad H = H_0 + V.} \tag{1.36}$$

For an electron of charge $q = -e$, with $e > 0$, relative to a fixed positive core, the electric dipole is $\boldsymbol{\mu}^{(e)} = \boldsymbol{p} = -e\boldsymbol{r}$; it points from negative to positive charge. Finally, we quantize the particle by replacing position and canonical momentum with operators, $\boldsymbol{r} \to \hat{\boldsymbol{r}}$ and $\boldsymbol{\pi} \to -i\hbar\boldsymbol{\nabla}$. In the electric-dipole approximation used for our optical transition, we retain the electric term and neglect the magnetic one[5]. The Hamiltonian is therefore

[5] *magnetic forces are much smaller than electrical ones in the case of an electron*

$$\boxed{\hat{H}(t) = \hat{H}_0 + \hat{V}(t), \qquad \hat{V}(t) = -\hat{\boldsymbol{p}} \cdot \boldsymbol{E}(t),} \tag{1.37}$$

with $\hat{\boldsymbol{p}} = -e\hat{\boldsymbol{r}}$ and the applied field evaluated at the atomic center. Note that this is a semiclassical Hamiltonian: the atom is quantum mechanical and the applied field remains classical.

1.4 Two-level atom Hamiltonian with EMF interaction

As an example, let us consider the two-level toy model introduced previously. Projecting the dipole onto the two retained states gives

$$\hat{\boldsymbol{p}} = \sum_{m,n=1}^{2} \boldsymbol{p}_{mn} |m\rangle\langle n|, \qquad \boldsymbol{p}_{mn} = \langle m|\hat{\boldsymbol{p}}|n\rangle. \tag{1.38}$$

Assuming spatial inversion symmetry and thus that the eigenstates have definite parity, there is no permanent dipole as discussed in previous week, i.e. $\boldsymbol{p}_{11} = \boldsymbol{p}_{22} = 0$. For a linearly polarized field $\boldsymbol{E}(t) = E(t)\boldsymbol{e}_p$, define the projected operator $\hat{p} = \boldsymbol{e}_p \cdot \hat{\boldsymbol{p}}$. Hermiticity gives $p_{21} = p_{12}^*$, and the relative phase of the two states may be chosen so that $p_{12} = p_{21} = d > 0$. Then with the midpoint energy origin, it is trivial to obtain

$$\boxed{\hat{H}'(t) = \frac{\hbar\omega_0}{2}\hat{\sigma}_z - dE(t)\hat{\sigma}_x} \longleftrightarrow \begin{pmatrix} -\hbar\omega_0/2 & -dE(t) \\ -dE(t) & \hbar\omega_0/2 \end{pmatrix}. \tag{1.39}$$

The off-diagonal field coupling will be responsible of creating coherence and transferring population. But first we need to understand how it evolves in time.

2 From stationary states to time-evolution

The Schrödinger equation introduced last week suggests that the quantum state evolves in time. Assuming that we have a closed system, we can argue that this motion is associated with a *unitary transformation*. More precisely, for every pair of times t_0 and t there exists an operator $\hat{U}(t, t_0)$ such that

Definition. Time-evolution operator.

$$\boxed{|\psi(t)\rangle = \hat{U}(t, t_0)|\psi(t_0)\rangle.} \tag{2.1}$$

Note that $\hat{U}(t, t_0)$ belongs to the physical system, meaning that the same operator evolves *every* possible state in the Hilbert space from t_0 to t. It is easy to define also some properties:

- For a closed system the **time-evolution operator is unitary**, i.e.

$$\boxed{\hat{U}^\dagger(t, t_0)\hat{U}(t, t_0) = \hat{\mathbb{I}},} \tag{2.2}$$

  as we need the normalization[6] to be preserved by time evolution, i.e.

$$\langle\psi(t)|\psi(t)\rangle = \langle\psi(t_0)|\hat{U}^\dagger(t, t_0)\hat{U}(t, t_0)|\psi(t_0)\rangle = \langle\psi(t_0)|\psi(t_0)\rangle. \tag{2.3}$$

[6] *and also any inner product*

- Evolving a state from a time t_0 to the same time cannot change it, i.e.

$$|\psi(t_0)\rangle = \hat{U}(t_0, t_0)|\psi(t_0)\rangle \Rightarrow \hat{U}(t_0, t_0) = \hat{\mathbb{I}}. \tag{2.4}$$

- Evolution over a finite interval may be divided into successive steps. If $t_0 < t_1 < t_2$, we have then

$$|\psi(t_2)\rangle = \hat{U}(t_2, t_1)\hat{U}(t_1, t_0)|\psi(t_0)\rangle, \tag{2.5}$$

  which gives the **composition law**

$$\boxed{\hat{U}(t_2, t_0) = \hat{U}(t_2, t_1)\hat{U}(t_1, t_0).} \tag{2.6}$$

  Note, however, that **the order of the operators matters**.

- **Evolution must be reversible for a closed system** which means that taking $t_2 = t_0$ in Eq. 2.6,

$$\hat{\mathbb{I}} = \hat{U}(t_0, t)\hat{U}(t, t_0). \Rightarrow \boxed{\hat{U}(t_0, t) = \hat{U}^{-1}(t, t_0) = \hat{U}^\dagger(t, t_0).} \tag{2.7}$$

All in all, for a closed quantum system, the time evolution should be a reversible rotation in Hilbert space. The state may change its amplitudes and relative phases, but a unitary evolution cannot change its norm or destroy information about the initial state.

2.1 Connection with the Schrödinger Equation

In the previous week, we stated that in the quantum mechanics framework, the dynamics of a closed quantum system can be described by a state that evolves according to the time-dependent Schrödinger equation. However, it is useful to take a step back and see how this equation is intrinsically connected with the existence of a time-evolution operator. Starting from

$$|\psi(t)\rangle = \hat{U}(t, t_0)|\psi(t_0)\rangle, \tag{2.8}$$

and differentiate with respect to t, we obtain

$$\frac{\partial}{\partial t}|\psi(t)\rangle = \frac{\partial \hat{U}(t, t_0)}{\partial t}|\psi(t_0)\rangle. \tag{2.9}$$

Noting that

$$|\psi(t_0)\rangle = \hat{U}^\dagger(t, t_0)|\psi(t)\rangle, \tag{2.10}$$

we obtain

$$\frac{\partial}{\partial t}|\psi(t)\rangle = \left[\frac{\partial \hat{U}(t, t_0)}{\partial t}\hat{U}^\dagger(t, t_0)\right]|\psi(t)\rangle. \tag{2.11}$$

As the operator in brackets is anti-Hermitian[7] we can define a Hermitian operator $\hat{H}(t)$ and write the generator of the evolution as

$$\frac{\partial \hat{U}}{\partial t}\hat{U}^\dagger = -\frac{i}{\hbar}\hat{H}(t). \tag{2.13}$$

Equation 2.11 then becomes

$$\boxed{i\hbar\frac{\partial}{\partial t}|\psi(t)\rangle = \hat{H}(t)|\psi(t)\rangle,} \tag{2.14}$$

which is exactly the time-dependent Schrödinger equation.

In optical problems concerning resonant interaction, it is useful to divide the Hamiltonian into a static part and an interaction,

$$\boxed{\hat{H}(t) = \hat{H}_0 + \hat{V}(t).} \tag{2.15}$$

The unperturbed Hamiltonian $\hat{H}_0$ describes the isolated material system,

[7] *follows directly by differentiating the unitarity condition which gives*

$$\frac{\partial \hat{U}}{\partial t}\hat{U}^\dagger + \hat{U}\frac{\partial \hat{U}^\dagger}{\partial t} = 0. \tag{2.12}$$

defining energy eigenstates that satisfy

$$\hat{H}_0|n\rangle = E_n|n\rangle = \hbar\omega_n|n\rangle. \tag{2.16}$$

The interaction $\hat{V}(t)$ accounts for whatever external influence drives the system. For us this is the electromagnetic field, through the electric-dipole interaction derived in Sec. 1.3.

If the states $|n\rangle$ form a complete orthonormal basis,

$$\langle m|n\rangle = \delta_{mn}, \qquad \sum_n |n\rangle\langle n| = \hat{\mathbb{I}}, \tag{2.17}$$

then any state can be expanded as

$$\boxed{|\psi(t)\rangle = \sum_n c_n(t)|n\rangle.} \tag{2.18}$$

The coefficients are probability amplitudes, together with a normalization condition

$$P_n(t) = |c_n(t)|^2 \Rightarrow \sum_n |c_n(t)|^2 = 1. \tag{2.19}$$

A point worth keeping in mind is that the choice of basis and the physical state are different things: the basis $\{|n\rangle\}$ is just a coordinate system in Hilbert space, whereas the state is the vector that is represented in that coordinate system.

2.2 Free evolution

Quantum dynamics essentially consists of finding how the expansion coefficients change with time in a basis that makes the physics more transparent or tractable. Let us first remove the external interaction ($\hat{V} = 0$) and take $\hat{H}_0$ independent of time. We already saw that if the system is initially in one energy eigenstate, then we have

$$|\psi(0)\rangle = |n\rangle \Rightarrow |\psi(t)\rangle = e^{-iE_nt/\hbar}|n\rangle = e^{-i\omega_n t}|n\rangle, \tag{2.20}$$

but let us obtain the same structure with the time evolution approach. Substituting the time evolution operator in the Schrödinger equation leads to an ordinary constant-coefficient differential equation,

$$\frac{d\hat{U}}{dt} = -\frac{i}{\hbar}\hat{H}\hat{U}, \tag{2.21}$$

whose formal solution is given by [8]

[8] *The exponential of an operator is defined by the usual power series,*

$$e^{\hat{A}} = \hat{\mathbb{I}} + \hat{A} + \frac{\hat{A}^2}{2!} + \frac{\hat{A}^3}{3!} + \cdots. \tag{2.22}$$

Evolution under a time-independent Hamiltonian:

$$\boxed{\hat{U}(t,t_0) = \exp\left[-\frac{i}{\hbar}\hat{H}(t-t_0)\right].} \tag{2.23}$$

Now, if we consider the initial state expanded in eigenstates of $\hat{H}$ as

$$|\psi(t_0)\rangle = \sum_n c_n |n\rangle, \tag{2.24}$$

with

$$\hat{H}|n\rangle = E_n|n\rangle, \tag{2.25}$$

then

$$|\psi(t)\rangle = e^{-i\hat{H}(t-t_0)/\hbar}\sum_n c_n|n\rangle \tag{2.26}$$

$$= \sum_n c_n e^{-iE_n(t-t_0)/\hbar}|n\rangle, \tag{2.27}$$

as obtained in previous week.

2.3 Time-dependent Hamiltonians

If the Hamiltonian depends explicitly on time,

$$\hat{H} = \hat{H}(t), \tag{2.28}$$

the evolution operator still obeys

$$i\hbar\frac{\partial}{\partial t}\hat{U}(t,t_0) = \hat{H}(t)\hat{U}(t,t_0). \tag{2.29}$$

Formal integration from t_0 to t gives

$$\hat{U}(t,t_0) = \hat{\mathbb{I}} - \frac{i}{\hbar}\int_{t_0}^{t}\hat{H}(t_1)\hat{U}(t_1,t_0)\,dt_1. \tag{2.30}$$

This equation is exact, but the unknown propagator still appears inside the integral. We may therefore substitute the equation into itself iteratively. The first iteration gives

$$\hat{U}(t,t_0) = \hat{\mathbb{I}} - \frac{i}{\hbar}\int_{t_0}^{t} dt_1\,\hat{H}(t_1) + \left(-\frac{i}{\hbar}\right)^2\int_{t_0}^{t}dt_1\int_{t_0}^{t_1}dt_2\,\hat{H}(t_1)\hat{H}(t_2) + \cdots. \tag{2.31}$$

This is the Dyson series and we shall notice the nested integration limits,

$$t_0 \leq t_2 \leq t_1 \leq t. \tag{2.32}$$

These ensure that the operator associated with the later time appears to the left and therefore acts later in the chronological sequence. This structure is written compactly using the time-ordering operator $\mathcal{T}$,

Time-ordered evolution operator:

$$\boxed{\hat{U}(t,t_0) = \mathcal{T}\exp\left[-\frac{i}{\hbar}\int_{t_0}^{t}\hat{H}(t')\,dt'\right].} \tag{2.33}$$

The symbol $\mathcal{T}$ instructs us to order products so that operators evaluated at later times appear to the left. If

$$[\hat{H}(t_1),\hat{H}(t_2)] = 0 \qquad \text{for all } t_1, t_2, \tag{2.34}$$

then time ordering becomes unnecessary and the familiar exponential of the integrated Hamiltonian is valid,

$$\hat{U}(t,t_0) = \exp\left[-\frac{i}{\hbar}\int_{t_0}^{t}\hat{H}(t')\,dt'\right]. \tag{2.35}$$

Yet this is usually not the case for time dependent Hamiltonians, as distinct time will correspond to distinct states. For the general case,

$$[\hat{H}(t_1),\hat{H}(t_2)] \neq 0, \tag{2.36}$$

the order in which the Hamiltonian acts becomes physically relevant. The solution must then keep track of this temporal ordering.

3 Perturbation theory

The time-ordered propagator is exact but does not by itself provide an easy calculation. One way to try to tackle this challenge and get the dynamics is to follow perturbation theory. We start by reviewing the stationary perturbation theory before introducing a dynamical version that will be useful for quantum optics.

3.1 Stationary perturbation theory

For a time-independent perturbation write $\hat{H} = \hat{H}_0 + \lambda\hat{V}$, where λ records the order and is set to one at the end. Let $\hat{H}_0|n\rangle = E_n|n\rangle$, with a nondegenerate

level n. We may then expand

$$\mathcal{E}_n = E_n + \lambda E_n^{(1)} + \lambda^2 E_n^{(2)} + \cdots \tag{3.1}$$

$$|\widetilde{n}\rangle = |n\rangle + \lambda |n^{(1)}\rangle + \cdots \tag{3.2}$$

and insert these expansions into $\hat{H}|\widetilde{n}\rangle = \mathcal{E}_n|\widetilde{n}\rangle$. Comparing first-order terms yields

$$(\hat{H}_0 - E_n)|n^{(1)}\rangle = -(\hat{V} - E_n^{(1)})|n\rangle. \tag{3.3}$$

Projection onto $\langle n|$ gives $E_n^{(1)} = V_{nn}$; projection onto $\langle m|$, $m \neq n$, gives $\langle m|n^{(1)}\rangle = V_{mn}/(E_n - E_m)$. Choosing $\langle n|n^{(1)}\rangle = 0$, and projecting the second-order equation onto $\langle n|$, we obtain

$$\boxed{\begin{aligned} \mathcal{E}_n &= E_n + V_{nn} + \sum_{m\neq n} \frac{|V_{mn}|^2}{E_n - E_m} + \cdots, \\ |\widetilde{n}\rangle &= |n\rangle + \sum_{m\neq n} \frac{V_{mn}}{E_n - E_m}|m\rangle + \cdots. \end{aligned}} \tag{3.4}$$

The displayed state is normalized to first order; normalization changes enter at second order. Small mixing requires $|V_{mn}|/|E_n - E_m| \ll 1$. Degenerate or nearly degenerate levels must instead be treated together by diagonalizing the perturbation in their subspace.

Example: DC Stark shift and induced dipole

We consider a static field $E(t) = E_{\mathrm{dc}}$ along the polarization direction already introduced. Then $\hat{V} = -dE_{\mathrm{dc}}\hat{\sigma}_x$, so the diagonal matrix elements vanish. From perturbation theory it is trivial to get the leading energy shifts as

$$\boxed{\delta E_1 = -\frac{d^2 E_{\mathrm{dc}}^2}{\hbar\omega_0}, \qquad \delta E_2 = +\frac{d^2 E_{\mathrm{dc}}^2}{\hbar\omega_0}.} \tag{3.5}$$

The lower level moves down and the upper level moves up. The transition frequency therefore increases by

$$\delta\omega_{21} = \frac{\delta E_2 - \delta E_1}{\hbar} = \frac{2d^2 E_{\mathrm{dc}}^2}{\hbar^2\omega_0}. \tag{3.6}$$

The corresponding states, to first order in $\eta = dE_{\mathrm{dc}}/(\hbar\omega_0)$, are

$$|\widetilde{1}\rangle = |1\rangle + \eta|2\rangle + O(\eta^2), \qquad |\widetilde{2}\rangle = |2\rangle - \eta|1\rangle + O(\eta^2). \tag{3.7}$$

Note that although the unperturbed states have zero dipole, their mix-

ture does not:

$$\boxed{\langle\widetilde{1}|\hat{p}|\widetilde{1}\rangle = \frac{2d^2}{\hbar\omega_0}E_{\mathrm{dc}} + O(E_{\mathrm{dc}}^3) \equiv \alpha(0)E_{\mathrm{dc}} + O(E_{\mathrm{dc}}^3).} \tag{3.8}$$

The upper state has the opposite induced dipole. Thus the ground-state polarizability is $\alpha(0) = 2d^2/(\hbar\omega_0)$ and $\delta E_1 = -\frac{1}{2}\alpha(0)E_{\mathrm{dc}}^2$. The same result follows from $\langle\hat{p}\rangle = -\partial\mathcal{E}_1/\partial E_{\mathrm{dc}}$. As a check, exact diagonalization of the midpoint Hamiltonian gives

$$\mathcal{E}_\pm = \pm\sqrt{(\hbar\omega_0/2)^2 + d^2E_{\mathrm{dc}}^2}, \tag{3.9}$$

whose expansion reproduces Eq. 3.5. Note that in this case, both the diagonalization and perturbation theory approaches are possible, yet, in systems with larger number of levels, perturbation theory is much simpler and analytically tractable.

3.2 Time-dependent perturbation theory

As previously said, the exact time-ordered evolution operator is formally complete but rarely easy to evaluate. Fortunately, many optical interactions begin in a regime where the applied field perturbs a system whose unperturbed states and energies are already known. In that situation we can organize the dynamics according to successive powers of the interaction strength. Again we write

$$\boxed{\hat{H}(t) = \hat{H}_0 + \lambda\hat{V}(t),} \tag{3.10}$$

where λ is an expansion parameter that keeps track of the perturbation order (which will be set equal to one at the end). Expanding the state in the basis of eigenstates of $\hat{H}_0$,

$$|\psi_S(t)\rangle = \sum_n C_n(t)e^{-i\omega_n t}|n\rangle. \tag{3.11}$$

the coefficients $C_n(t)$ are now time-dependent variables that are associated with changes produced by the interaction. Substituting Eq. 3.11 into the Schrödinger equation,

$$i\hbar\frac{d}{dt}|\psi\rangle = (\hat{H}_0 + \lambda\hat{V})|\psi\rangle \tag{3.12}$$

gives

$$i\hbar \sum_n \left[\dot{C}_n - i\omega_n C_n\right] e^{-i\omega_n t}|n\rangle = \sum_n C_n e^{-i\omega_n t}\hat{H}_0|n\rangle + \lambda \sum_n C_n e^{-i\omega_n t}\hat{V}(t)|n\rangle. \tag{3.13}$$

Using $\hat{H}_0|n\rangle = \hbar\omega_n|n\rangle$, the free-evolution terms cancel, and projecting from the left with $\langle m|$ gives

Coupled equations for the interaction amplitudes:

$$\boxed{\dot{C}_m(t) = -\frac{i\lambda}{\hbar}\sum_n V_{mn}(t)e^{i\omega_{mn}t}C_n(t),} \tag{3.14}$$

where

$$V_{mn}(t) = \langle m|\hat{V}(t)|n\rangle \tag{3.15}$$

and

$$\omega_{mn} = \omega_m - \omega_n = \frac{E_m - E_n}{\hbar}. \tag{3.16}$$

This is a central result that tells us that a perturbation can transfer amplitude between states through its matrix elements, and that each coupling term is accompanied by the relative phase accumulated between the two unperturbed states. Note that this also connects directly to the selection rules discussed in Week II. If symmetry forces

$$V_{mn} = 0, \tag{3.17}$$

then no first-order coupling exists between those states, regardless of how closely the driving frequency matches their energy separation.

3.2.1 Perturbative expansion

The coupled system of equations only gives us a formalism, but the equations are still dependent. To tackle them in perturbative manner, we expand

$$C_n(t) = C_n^{(0)}(t) + \lambda C_n^{(1)}(t) + \lambda^2 C_n^{(2)}(t) + \cdots. \tag{3.18}$$

Substituting into Eq. 3.14 and comparing equal powers of λ yields a hierarchy.

- At **zeroth order**,

$$\dot{C}_m^{(0)} = 0. \tag{3.19}$$

  Thus the unperturbed coefficients are constant.

- At **first order**,

$$\dot{C}_m^{(1)}(t) = -\frac{i}{\hbar}\sum_n V_{mn}(t)e^{i\omega_{mn}t}C_n^{(0)}. \tag{3.20}$$

- At **second order**,

$$\dot{C}_m^{(2)}(t) = -\frac{i}{\hbar}\sum_n V_{mn}(t)e^{i\omega_{mn}t}C_n^{(1)}(t). \tag{3.21}$$

meaning that every new order is generated by the interaction acting on the result obtained at the preceding order.

For now we will focus on the first-order transition amplitude. Suppose the system is prepared at time t_0 in a single eigenstate $|i\rangle$. Then

$$C_i^{(0)} = 1, \qquad C_{n\neq i}^{(0)} = 0. \tag{3.22}$$

For a different final state $|f\rangle$, Eq. 3.20 becomes

$$\dot{C}_f^{(1)}(t) = -\frac{i}{\hbar}V_{fi}(t)e^{i\omega_{fi}t}. \tag{3.23}$$

Integrating from t_0 to t gives

First-order transition amplitude:

$$\boxed{C_f^{(1)}(t) = -\frac{i}{\hbar}\int_{t_0}^{t} V_{fi}(t')e^{i\omega_{fi}t'}\,dt'.} \tag{3.24}$$

This equation means that the transition amplitude is the Fourier overlap between the time-dependent perturbation and the natural phase evolution of the transition. If the perturbation oscillates rapidly with a frequency far from ω_{fi}, positive and negative contributions tend to cancel. If the perturbation contains a component close to the transition frequency, the phase varies slowly and the contributions can accumulate coherently.

Fermi Golden Rule: Within first-order perturbation theory, and assuming a constant matrix element $V_{fi}(t') = V_{fi} = \langle f|\hat{V}|i\rangle$ turned on at $t_0 = 0$:

$$C_f^{(1)}(t) = -\frac{i}{\hbar}V_{fi}\int_0^t e^{i\omega_{fi}t'}\,dt' = -\frac{V_{fi}}{\hbar\omega_{fi}}\left(e^{i\omega_{fi}t} - 1\right) \tag{3.25}$$

The transition probability $P_{i\to f}(t) = |C_f^{(1)}(t)|^2$ is

$$P_{i\to f}(t) = \frac{|V_{fi}|^2}{\hbar^2\omega_{fi}^2}\left|e^{i\omega_{fi}t} - 1\right|^2 = \frac{|V_{fi}|^2}{\hbar^2}\frac{4\sin^2\left(\frac{\omega_{fi}t}{2}\right)}{\omega_{fi}^2} \tag{3.26}$$

Using the representation of the Dirac delta function $\lim_{t\to\infty}\frac{\sin^2(\alpha t)}{\pi t\alpha^2} = \delta(\alpha)$ and expressing in terms of the energy difference we have

$$\lim_{t\to\infty}\frac{\sin^2\left(\frac{\Delta E\cdot t}{2\hbar}\right)}{\left(\frac{\Delta E}{2\hbar}\right)^2} = \pi t\,\delta\left(\frac{\Delta E}{2\hbar}\right) = 2\pi\hbar t\,\delta(\Delta E) \tag{3.27}$$

which yields the linear-in-time probability

$$P_{i\to f}(t) = \frac{2\pi}{\hbar}|V_{fi}|^2\delta(E_f - E_i)\,t \tag{3.28}$$

The transition rate per unit time $\Gamma_{i\to f} = \frac{dP_{i\to f}}{dt}$ is

$$\Gamma_{i\to f} = \frac{2\pi}{\hbar}|V_{fi}|^2\delta(E_f - E_i) \tag{3.29}$$

or integrating over a continuum of final states with density of states $\rho(E_f)$,

$$\Gamma_{i\to f} = \int\frac{2\pi}{\hbar}|V_{fi}|^2\delta(E_f - E_i)\rho(E_f)\,dE_f \tag{3.30}$$

corresponding to the Fermi's Golden Rule.

At the second order, the structure becomes richer. Substituting the first-order solution into Eq. 3.21 gives

$$C_f^{(2)}(t) = \left(-\frac{i}{\hbar}\right)^2\sum_n\int_{t_0}^{t}dt_1\int_{t_0}^{t_1}dt_2\,V_{fn}(t_1)V_{ni}(t_2) \times e^{i\omega_{fn}t_1}e^{i\omega_{ni}t_2}. \tag{3.31}$$

The sum thus runs over intermediate states $|n\rangle$. The perturbation first transfers amplitude from $|i\rangle$ to an intermediate state and then, at a later time, from that intermediate state to $|f\rangle$. This is similar to what happens in nonlinear optics when we generate new frequencies.

3.3 Example: a two-level atom driven by light

Consider again our two-level toy model but now take $E(t) = E_0 \cos\omega t$ for $t \geq 0$, and prepare $|\psi(0)\rangle = |1\rangle$. Defining

$$\Omega = \frac{dE_0}{\hbar}, \qquad \Delta = \omega_0 - \omega, \qquad \Sigma = \omega_0 + \omega, \tag{3.32}$$

where E_0 is the real peak field, Ω is the resonant Rabi frequency, Δ is a detuning, with positive detuning meaning light below the atomic resonance, with $C_1^{(0)} = 1$ and $C_2^{(0)} = 0$, Eq. 3.24 gives

$$C_2^{(1)}(t) = i\Omega \int_0^t \cos(\omega t') e^{i\omega_0 t'}\, dt' = \frac{\Omega}{2}\left[\frac{e^{i\Delta t} - 1}{\Delta} + \frac{e^{i\Sigma t} - 1}{\Sigma}\right]. \tag{3.33}$$

The denominators show two distinct scales: a sufficient weak-excitation condition away from resonance is $|\Omega/\Delta| \ll 1$ and $|\Omega/\Sigma| \ll 1$, corresponding to a "large detuning" situation [9]. Note however, that over very long times, small higher-order energy shifts can still accumulate phase errors in the transient solution. Using the dipole operator it is easy to show that

[9] *i.e. it need not be large compared with the optical frequency* ω_0

$$\langle \hat{p}(t) \rangle = 2d\ \mathrm{Re}\left[C_1^*(t) C_2(t) e^{-i\omega_0 t}\right], \tag{3.34}$$

finding that to first order in the applied field

$$\boxed{\langle \hat{p}(t) \rangle^{(1)} = \frac{d^2 E_0}{\hbar}\left(\frac{1}{\Delta} + \frac{1}{\Sigma}\right)[\cos\omega t - \cos\omega_0 t].} \tag{3.35}$$

To relate with the optical response, we now consider a dilute ensemble with number density $\mathcal{N}$, identical transition projections d, and all atoms initially in the lower state, $P = \mathcal{N}\langle \hat{p} \rangle$. We neglect local-field corrections. Focusing on the forced component of the dipole [10], gives

[10] *and preparing it by an adiabatically switched weak field away from resonance*

$$\boxed{\begin{aligned}\chi^{(1)}(\omega) &= \frac{\mathcal{N} d^2}{\epsilon_0 \hbar}\left(\frac{1}{\omega_0 - \omega} + \frac{1}{\omega_0 + \omega}\right) \\ &= \frac{2\mathcal{N} d^2 \omega_0}{\epsilon_0 \hbar(\omega_0^2 - \omega^2)}.\end{aligned}} \tag{3.36}$$

Note that an isotropic orientational average replaces the squared projected dipole by the appropriate average, $|\boldsymbol{p}_{12}|^2/3$ for randomly oriented identical dipoles.

To compare, with the classical model, we consider the Lorentz model for

the susceptibility derived last week,

$$\chi_{\mathrm{L}}(\omega) = \frac{\mathcal{N}e^2}{\epsilon_0 m(\omega_0^2 - \omega^2)}. \tag{3.37}$$

For the undamped case, i.e. $\gamma = 0$, the frequency dependence is identical. Defining the oscillator strength for this polarization by

$$f_{12} = \frac{2m\omega_0 d^2}{\hbar e^2}, \qquad \chi^{(1)} = f_{12}\chi_{\mathrm{L}}, \tag{3.38}$$

identifies how the quantum transition matrix element sets the classical oscillator weight, solving one problem of the classical model - the need for empirical assumptions. Yet, at exact resonance the continuous limit of Eq. 3.33 is

$$C_2^{(1)}(t) = \frac{i\Omega t}{2} + \frac{\Omega}{4\omega_0}(e^{2i\omega_0 t} - 1). \tag{3.39}$$

The resonant term thus grows with time, so perturbation theory requires $\Omega t \ll 1$. After averaging the fast optical correction, it predicts $P_2 \simeq \Omega^2 t^2/4$, which cannot remain valid once it is appreciable. Thus a weak drive may still require a nonperturbative solution at sufficiently long times. The undamped frequency-domain formula is also real away from resonance and has no finite linewidth, so we need to introduce a distinct strategy.

4 Pictures of quantum dynamics

So far we assigned the time dependence to the state. But quantum mechanics does not require this explicitly. Indeed, we can either assign temporal dependence to state $|\psi\rangle$, to the operator $\hat{O}$, or both. Different choices only correspond to different coordinate descriptions of the same physical evolution, but may strongly simplify the calculations involved in quantum dynamics description. We call these choices **pictures**, which typically remove evolution we already understand and leave the part of the dynamics we actually want to calculate.

4.1 Schrödinger picture

In the Schrödinger picture the states evolve according to

$$i\hbar\frac{d}{dt}|\psi_S(t)\rangle = \hat{H}_S(t)|\psi_S(t)\rangle, \tag{4.1}$$

while an operator with no explicit time dependence remains fixed,

$$\hat{O}_S(t) = \hat{O}_S. \tag{4.2}$$

The observable is thus computed from

$$\langle \hat{O} \rangle = \langle \psi_S(t) | \hat{O}_S | \psi_S(t) \rangle. \tag{4.3}$$

This is the picture used in most introductory quantum mechanics, where the state vector carries the complete dynamical evolution.

4.2 Heisenberg picture

The Heisenberg picture transfers the time dependence from the state to the operators completely. Taking $t_0 = 0$ for simplicity, we have

$$|\psi_H\rangle = |\psi_S(0)\rangle, \tag{4.4}$$

while

$$\boxed{\hat{O}_H(t) = \hat{U}^\dagger(t) \hat{O}_S \hat{U}(t).} \tag{4.5}$$

The expectation value is now written

$$\langle \hat{O} \rangle = \langle \psi_H | \hat{O}_H(t) | \psi_H \rangle. \tag{4.6}$$

Differentiating Eq. 4.5 yields an equation of motion for the evolution of the operator

Heisenberg equation of motion:

$$\boxed{\frac{d\hat{O}_H}{dt} = \frac{i}{\hbar}[\hat{H}_H, \hat{O}_H] + \left(\frac{\partial \hat{O}}{\partial t}\right)_H.} \tag{4.7}$$

or for an operator with no explicit time dependence,

$$\frac{d\hat{O}_H}{dt} = \frac{i}{\hbar}[\hat{H}_H, \hat{O}_H]. \tag{4.8}$$

Provided that $\hat{O}$ has no explicit time dependence, the commutator plays the role of a dynamical derivative. In specific, if it commutes with the Hamiltonian, the observable is a constant of motion

$$[\hat{H}, \hat{O}] = 0 \Rightarrow \frac{d\hat{O}_H}{dt} = 0. \tag{4.9}$$

The Heisenberg picture is particularly interesting to analyze constants of motion, however in the presence of interaction terms, the need to compute the explicit form of the commutator can be challenging to keep track of all the time-dependent terms.

Example 1. For a free particle,

$$\hat{H} = \frac{\hat{p}^2}{2m}.$$

Using $[\hat{x}, \hat{p}] = i\hbar$,

$$\frac{d\hat{x}_H}{dt} = \frac{i}{\hbar}[\hat{H}, \hat{x}_H] = \frac{\hat{p}_H}{m},$$

which has exactly the form of the classical velocity relation. Likewise,

$$\frac{d\hat{p}_H}{dt} = 0.$$

The Heisenberg picture therefore makes the connection between operator dynamics and classical equations of motion particularly transparent.

4.3 Interaction picture

Alternatively to the Schrödinger and Heisenberg pictures, the interaction picture is a representation that is especially important for driven optical systems. Suppose we have

$$\hat{H}(t) = \hat{H}_0 + \hat{V}(t), \tag{4.10}$$

and the free evolution generated by $\hat{H}_0$ is known and easy to obtain. Then we define

$$\hat{U}_0(t) = e^{-i\hat{H}_0 t/\hbar} \tag{4.11}$$

and transform the Schrödinger-picture state according to this operator as

$$\boxed{|\psi_I(t)\rangle = \hat{U}_0^\dagger(t)|\psi_S(t)\rangle.} \tag{4.12}$$

Differentiating this expression gives

$$i\hbar\frac{d}{dt}|\psi_S\rangle = \hat{H}_0\hat{U}_0|\psi_I\rangle + i\hbar\hat{U}_0\frac{d}{dt}|\psi_I\rangle. \tag{4.13}$$

But the full Schrödinger equation also gives

$$i\hbar\frac{d}{dt}|\psi_S\rangle = (\hat{H}_0 + \hat{V}_S)\hat{U}_0|\psi_I\rangle. \tag{4.14}$$

The $\hat{H}_0$ terms cancel, leaving

$$i\hbar\hat{U}_0\frac{d}{dt}|\psi_I\rangle = \hat{V}_S\hat{U}_0|\psi_I\rangle. \tag{4.15}$$

Multiplying by $\hat{U}_0^\dagger$ gives

Interaction-picture equation of motion:

$$\boxed{i\hbar\frac{d}{dt}|\psi_I(t)\rangle = \hat{V}_I(t)|\psi_I(t)\rangle,} \tag{4.16}$$

where

$$\boxed{\hat{V}_I(t) = \hat{U}_0^\dagger(t)\hat{V}_S(t)\hat{U}_0(t) = e^{+i\hat{H}_0 t/\hbar}\hat{V}_S(t)e^{-i\hat{H}_0 t/\hbar}.} \tag{4.17}$$

Conceptually, this transformation allows us to remove from the state the rapid and exactly known evolution generated by $\hat{H}_0$, and focus only on the change caused by the interaction alone. In a typical quantum optics problem, this is often the part we want to follow. Note also that for an arbitrary operator, we have

$$\frac{d\hat{O}_I}{dt} = \frac{i}{\hbar}[\hat{H}_0, \hat{O}_I] + \left(\frac{\partial\hat{O}}{\partial t}\right)_I, \tag{4.18}$$

meaning that in the interaction picture the state evolves because of $\hat{V}_I$, while operators inherit the free evolution generated by $\hat{H}_0$.

5 Rotating Wave Approximation

For the two-level atom, $\hat{U}_0^\dagger\hat{\sigma}_\pm\hat{U}_0 = e^{\pm i\omega_0 t}\hat{\sigma}_\pm$. Consequently, the dipole operator becomes

$$\hat{p}_I(t) = d\left(\hat{\sigma}_- e^{-i\omega_0 t} + \hat{\sigma}_+ e^{i\omega_0 t}\right). \tag{5.1}$$

For $E(t) = E_0\cos\omega t$ the expansion of the terms yields

$$\hat{V}_I(t) = -\frac{\hbar\Omega}{2}\left[\hat{\sigma}_+(e^{i\Delta t} + e^{i\Sigma t}) + \hat{\sigma}_-(e^{-i\Delta t} + e^{-i\Sigma t})\right]. \tag{5.2}$$

This means that the change to the interaction picture separates terms rotating at the detuning Δ from those rotating at the much larger sum frequency Σ.

If $|\Delta|, \Omega \ll \Sigma \simeq 2\omega_0$, the terms at Σ average rapidly on the timescale of the population dynamics. Dropping them gives the **rotating-wave approximation** (RWA):

$$\boxed{\hat{V}_I^{\mathrm{RWA}}(t) = -\frac{\hbar\Omega}{2}\left(\hat{\sigma}_+ e^{i\Delta t} + \hat{\sigma}_- e^{-i\Delta t}\right).} \tag{5.3}$$

For $|\psi_I\rangle = C_1|1\rangle + C_2|2\rangle$, the coupled dynamical equations are

$$\dot{C}_1 = \frac{i\Omega}{2}e^{-i\Delta t}C_2, \qquad \dot{C}_2 = \frac{i\Omega}{2}e^{i\Delta t}C_1. \tag{5.4}$$

These can be solved even without setting $C_1 = 1$. In particular, the RWA does not require $\Omega \ll |\Delta|$ and can remain excellent on resonance even when first-order perturbation theory fails. It does still require the coupling to be small on the optical-frequency scale. Counter-rotating frequency shifts of order Ω^2/Σ and their accumulated long-time phase are omitted; due to this reason, the RWA is also inappropriate for obtaining the DC limit.

5.1 Exact solution within the RWA: Rabi oscillations

By introducing a rotating frame setting

$$C_1 = e^{-i\Delta t/2} b_1, \qquad C_2 = e^{i\Delta t/2} b_2, \tag{5.5}$$

the amplitudes $\boldsymbol{b} = (b_1, b_2)^T$ obey a time-independent equation,

$$i\hbar \dot{\boldsymbol{b}} = \hat{H}_{\text{rot}} \boldsymbol{b}, \qquad \hat{H}_{\text{rot}} = \frac{\hbar}{2} \begin{pmatrix} -\Delta & -\Omega \\ -\Omega & \Delta \end{pmatrix} = \frac{\hbar}{2} (\Delta \hat{\sigma}_z - \Omega \hat{\sigma}_x). \tag{5.6}$$

Since $\hat{H}^2_{\text{rot}} = (\hbar \Omega_R/2)^2 \hat{\mathbb{I}}$, where

$$\Omega_R = \sqrt{\Delta^2 + \Omega^2}, \tag{5.7}$$

and differentiating the Hamiltonian equation again,

$$i\hbar \ddot{\boldsymbol{b}} = \hat{H}_{\text{rot}} \dot{\boldsymbol{b}} \Rightarrow \ddot{\boldsymbol{b}} = -(\Omega_R/2)^2 \boldsymbol{b}, \tag{5.8}$$

which has the classical form of an harmonic oscillator differential equation, with an oscillation frequency Ω_R.

For an atom initially in $|1\rangle$ it can be shown[11] that

$$C_1(t) = e^{-i\Delta t/2} \left[\cos \frac{\Omega_R t}{2} + i \frac{\Delta}{\Omega_R} \sin \frac{\Omega_R t}{2} \right], \tag{5.9}$$

$$C_2(t) = i \frac{\Omega}{\Omega_R} e^{i\Delta t/2} \sin \frac{\Omega_R t}{2}. \tag{5.10}$$

[11] *applying initial conditions for* $b_1(0) = 1, b_2(0) = 0$, *for example*

Hence

$$\boxed{P_2(t) = \frac{\Omega^2}{\Omega_R^2} \sin^2 \frac{\Omega_R t}{2}, \qquad P_1(t) = 1 - P_2(t).} \tag{5.11}$$

On resonance, $P_2 = \sin^2(\Omega t/2)$: complete population transfer occurs at $t = \pi/\Omega$ and reverses coherently. At short times it agrees with $P_2 \simeq \Omega^2 t^2/4$. For $|\Delta| \gg \Omega$, the excitation remains small and the leading result is $(\Omega^2/\Delta^2) \sin^2(\Delta t/2)$, recovering near-resonant perturbation theory. None of these oscillations decays in the present model.

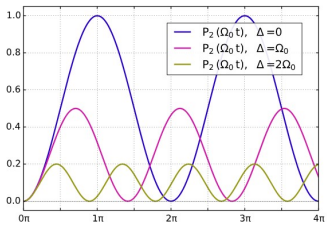


Figure 1. Rabi oscillations.

5.2 Dipole response and susceptibility without decay

To calculate the optical polarization we shall look at the coherence. Let $c = \cos(\Omega_R t/2)$ and $s = \sin(\Omega_R t/2)$. The Rabi amplitudes give

$$C_1^* C_2 e^{-i\omega_0 t} = e^{-i\omega t}\left[\frac{\Delta\Omega}{\Omega_R^2}s^2 + i\frac{\Omega}{\Omega_R}cs\right]. \tag{5.12}$$

Using Eq. 3.34, we obtain

$$\begin{aligned}\langle\hat{p}(t)\rangle = &\frac{d\Delta\Omega}{\Omega_R^2}[1-\cos(\Omega_R t)]\cos\omega t \\ &+ \frac{d\Omega}{\Omega_R}\sin(\Omega_R t)\sin\omega t.\end{aligned} \tag{5.13}$$

For the same dilute ensemble, it is easy to get

$$\chi_{\text{eff}} = \frac{\mathcal{N}d^2}{\epsilon_0\hbar}\left[\frac{\Delta}{\Omega_R^2}(1-\cos\Omega_R t) + \frac{i}{\Omega_R}\sin\Omega_R t\right]. \tag{5.14}$$

Note that this is an intensity-dependent, time-dependent response for a specified initial state and sudden turn-on, exact within the RWA. It is not a stationary linear susceptibility. Also, the imaginary part describes reversible energy exchange as well as excitation, meaning that it is not by itself a dissipative absorption coefficient. At fixed nonzero Δ, the weak-field limit reduces to

$$\chi_{\text{eff}}^{(1)}(t) = \frac{\mathcal{N}d^2}{\epsilon_0\hbar\Delta}(1-e^{-i\Delta t}). \tag{5.15}$$

or averaging over time,

$$\boxed{\chi_{\text{RWA}}^{(1)}(\omega) = \frac{\mathcal{N}d^2}{\epsilon_0\hbar(\omega_0-\omega)}.} \tag{5.16}$$

In the overlap regime $\Omega \ll |\Delta| \ll \omega_0$, this partially agrees with Eq. 3.36, since $1/\Sigma$ is then a small correction. The bounded Rabi dynamics resolves the unphysical long-time growth, but it does not produce a steady state or a finite resonance linewidth. However, far from the optical resonance or in the static limit, the full two-frequency result must be used instead, and the RWA regime fails. Relaxation and dephasing are the missing ingredients, and will be added with a density-matrix description of the next week.

Additional practice exercises

Exercise 1. A system is prepared in the state

$$|\psi(0)\rangle = \frac{1}{\sqrt{2}}\left(|1\rangle + e^{i\phi}|2\rangle\right).$$

Assuming free evolution under $\hat{H}_0$, calculate $|\psi(t)\rangle$ and the expectation value of an operator for which $O_{11} = O_{22} = 0$ and $O_{12} = O_{21} = O_0$. Show explicitly how the measured oscillation depends on the initial phase ϕ.

Exercise 2. Starting from

$$\hat{H}_0' = \frac{\hbar\omega_0}{2}\hat{\sigma}_z,$$

derive the free two-level propagator $\hat{U}_0(t)$. Apply it to the initial state $(|1\rangle + |2\rangle)/\sqrt{2}$ and verify that the populations remain fixed while the relative phase evolves.

Exercise 3. Derive the Heisenberg equation of motion by differentiating

$$\hat{O}_H(t) = \hat{U}^\dagger(t)\hat{O}_S\hat{U}(t)$$

for a time-independent Hamiltonian. Then apply it to the harmonic oscillator position and momentum operators and recover the corresponding classical equations of motion.

Exercise 4. Starting from

$$|\psi_S(t)\rangle = e^{-i\hat{H}_0 t/\hbar}|\psi_I(t)\rangle,$$

derive the interaction-picture equation

$$i\hbar\frac{d}{dt}|\psi_I(t)\rangle = \hat{V}_I(t)|\psi_I(t)\rangle.$$

State clearly which part of the dynamics has been removed by the transformation.

Exercise 5. A perturbation has a single matrix element

$$V_{fi}(t) = V_0 e^{-i\omega t}$$

for $0 < t < T$ and vanishes otherwise. Use first-order perturbation theory

to calculate $C_f^{(1)}(T)$. Show that the transition probability is peaked near $\omega = \omega_{fi}$ and identify the characteristic spectral width associated with a finite interaction time T.

Exercise 6. Consider two successive perturbations $\hat{V}_A$ and $\hat{V}_B$ represented by short pulses. Show to second order in the pulse strengths that reversing their order changes the final state by a term proportional to $[\hat{V}_B, \hat{V}_A]$. Interpret the result physically.

Exercise 7. Diagonalize the static two-level Hamiltonian in Eq. 1.39. Expand its eigenvalues to second order in E_{dc} and recover both the Stark shifts and the induced ground-state dipole from the energy derivative.

Exercise 8. Starting from Eq. 3.33, derive the switched-field dipole including its term at ω_0. Verify that the forced-response susceptibility has the DC limit obtained from the Stark shift. Explain why dropping the counter-rotating term spoils this check.

Exercise 9. Use the Rabi amplitudes to derive Eq. 5.13. Compare the resonant response at short and long times with first-order perturbation theory. Explain why the exact RWA result still does not define a unique stationary susceptibility without decay.